These op-amps are not matched for modern operational op-amps for high-quality audio applications. However, these are suitable for IP telephony and other low-quality audio applications.

This headphones buffer can drive headphones with resistance as low as 32-ohm using 5V power supply.

Circuit and working

The circuit diagram of the configurable audio buffer for headphones is shown in Fig. 1. The circuit works with both LMV358 and LM358 op-amps. The main differences are in voltage supply range, output current, output power and quiescent currents. To drive low-impedance headphones you may need a larger current source.

Table II gives minimal and typical output currents produced by a single op-amp and two LM358 op-amps in parallel. With these op-amps, there is enough current to drive even low-impedance headphones up to 32-ohm.

Connectors CON2 and CON3 are used for connecting audio inputs—CON3 can be used to connect with other audio equipment. Input resistance is set to 1-mega-ohm using resistors R7 and R8. This input resistance makes the circuit appropriate for many purposes, including direct connection to pickups of electrical guitars.

Outputs of this buffer are available across CON4 and CON5, which are connected in parallel. This means, the headphones can be used from CON4 and, at the same time, record music with a PC through CON5.

Buffer configurations

The audio buffer has the following configurations:

1. Stereo buffer with two inde-

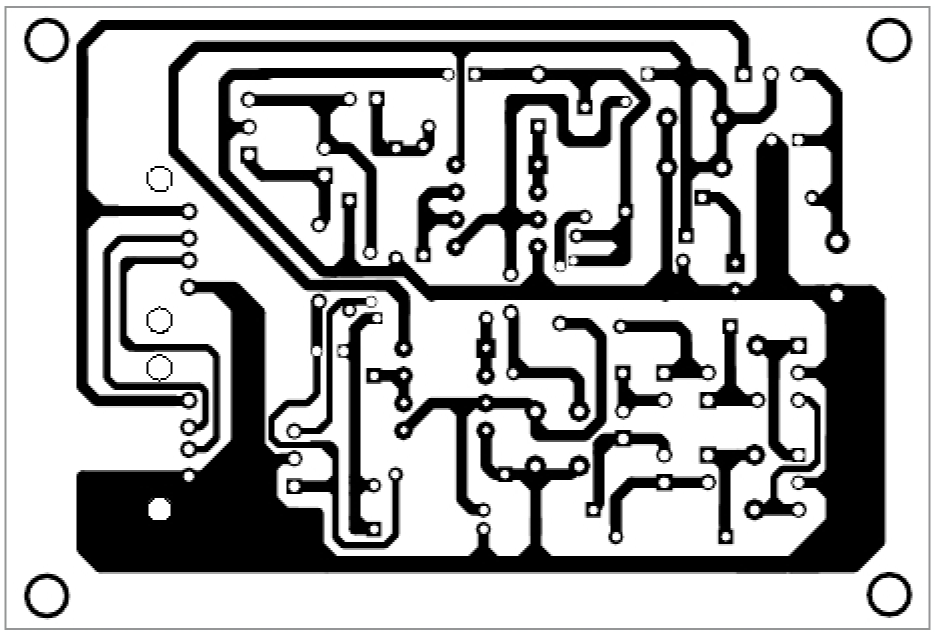

Fig. 2: Actual-size PCB layout of configurable audio buffer for headphones

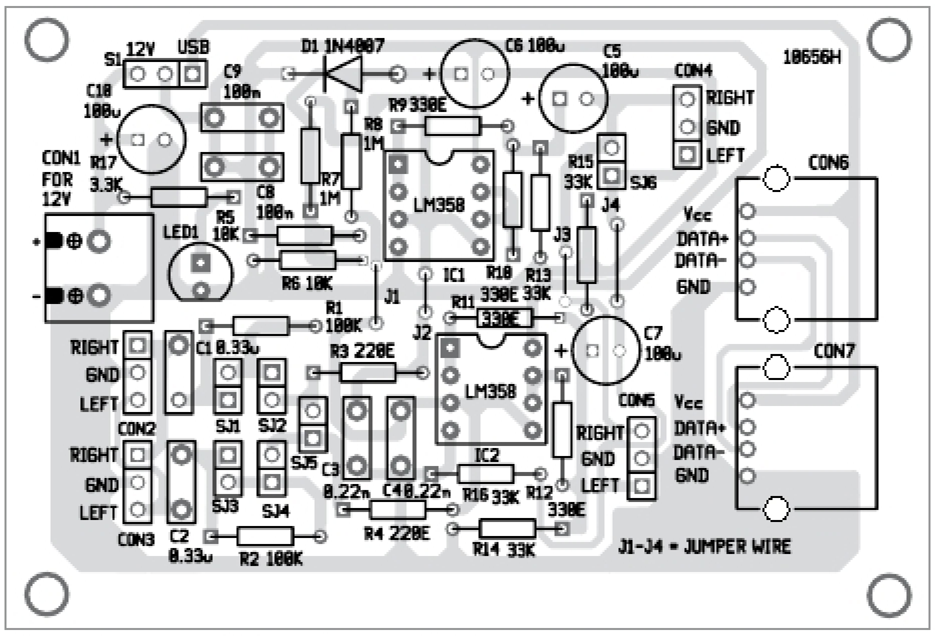

Fig. 3: Components layout for the PCB

TABLE II

SINK AND SOURCE CURRENTS OF AUDIO BUFFER FOR SINGLE AND TWO PARALLELLED LM358

Single op-amp				Two parallelled LM358 op-amps			
Sink		Source		Sink		Source	
Min	Typ	Min	Typ	Min	Typ	Min	Typ
10mA	20mA	20mA	40mA	20mA	40mA	40mA	80mA